NETWORK CABLE AND CONNECTORS

Course: Diploma in Cyber Security

Topic: Network Cable and Connectors

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INTRODUCTION

Network cables and connectors are used to connect computers, routers, switches and other network devices. Different cables are made for different speeds and distances. In this assignment, I have covered Ethernet cables, RJ-45 connectors, cable types, crimping steps and a simple gaming zone network setup.

Task 1: Cat6 Ethernet Cable and T568B Standard

A Cat6 Ethernet cable normally uses an RJ-45 connector. The T568B standard has a fixed wire colour order. The wire order from Pin 1 to Pin 8 is given below:

Pin	T568B Wire Colour	Pair / Use
1	White-Orange	Pair 2
2	Orange	Pair 2

3	White-Green	Pair 3
4	Blue	Pair 1
5	White-Blue	Pair 1
6	Green	Pair 3
7	White-Brown	Pair 4
8	Brown	Pair 4

FIGURE TO ADD HERE: Draw a labeled Cat6 cable with an RJ-45 connector. Show Pins 1-8 and write the T568B colour order next to the pins.

Figure Caption: Figure 1: Cat6 Ethernet Cable with RJ-45 Connector and T568B Pin Order

Task 2: Types of Network Cables

Cable Type	Example Device / Service	Where It Can Be Used	Why
Cat5e	Wi-Fi Router / PC	Home or small office network	It is suitable for normal Ethernet connections and is usually enough for basic internet and LAN use.
Cat6	Gaming PC / Zomato POS Terminal	Gaming zone, office or shop	It supports higher-speed Ethernet and gives a reliable wired

			connection.
Fiber Optic	IPL Streaming Server	Long-distance and high-speed links	It can carry large amounts of data over long distances with low signal loss.

Note: The real network of a service may use many different cable types, switches, routers and other technologies. The examples here are simple use cases for understanding where each cable can fit.

Task 3: RJ-45 Crimping Process

For this task, I studied the basic process of putting an RJ-45 connector on an Ethernet cable. The steps below are written in simple words.

1. Cut the Ethernet cable to the required length and remove a small part of the outer jacket.

2. Separate the four twisted pairs carefully and straighten the individual wires.

3. Arrange the eight wires in the T568B order: White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.

4. Cut the wire ends evenly so all eight wires are the same length.

5. Insert the wires fully into the RJ-45 connector and make sure each wire reaches its correct pin.

6. Put the connector into the crimping tool and press it firmly to lock the contacts in place.

7. Check the connector and, if available, use a cable tester to confirm the pin order and connection.

Two common mistakes to avoid:

- 1. Mixing up the T568B colour order.
- 2. Not pushing all eight wires fully into the RJ-45 connector before crimping.

Task 4: Cable Choice for IPL Screening Gaming Zone

For a gaming zone with multiple screens and consoles, I would prefer Cat6 Ethernet cable for the wired network. It gives a fast and stable connection and is a better choice for gaming than depending only on Wi-Fi. RJ-45 connectors can be used at the cable ends to connect PCs, consoles, switches and routers. For a very long connection between buildings or to a high-speed backbone, fiber optic cable can be used.

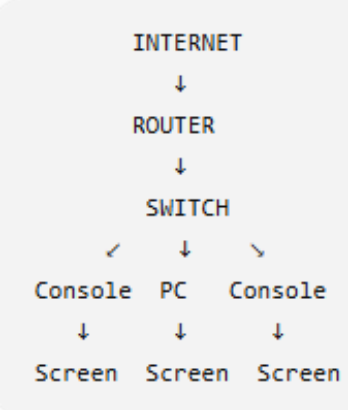
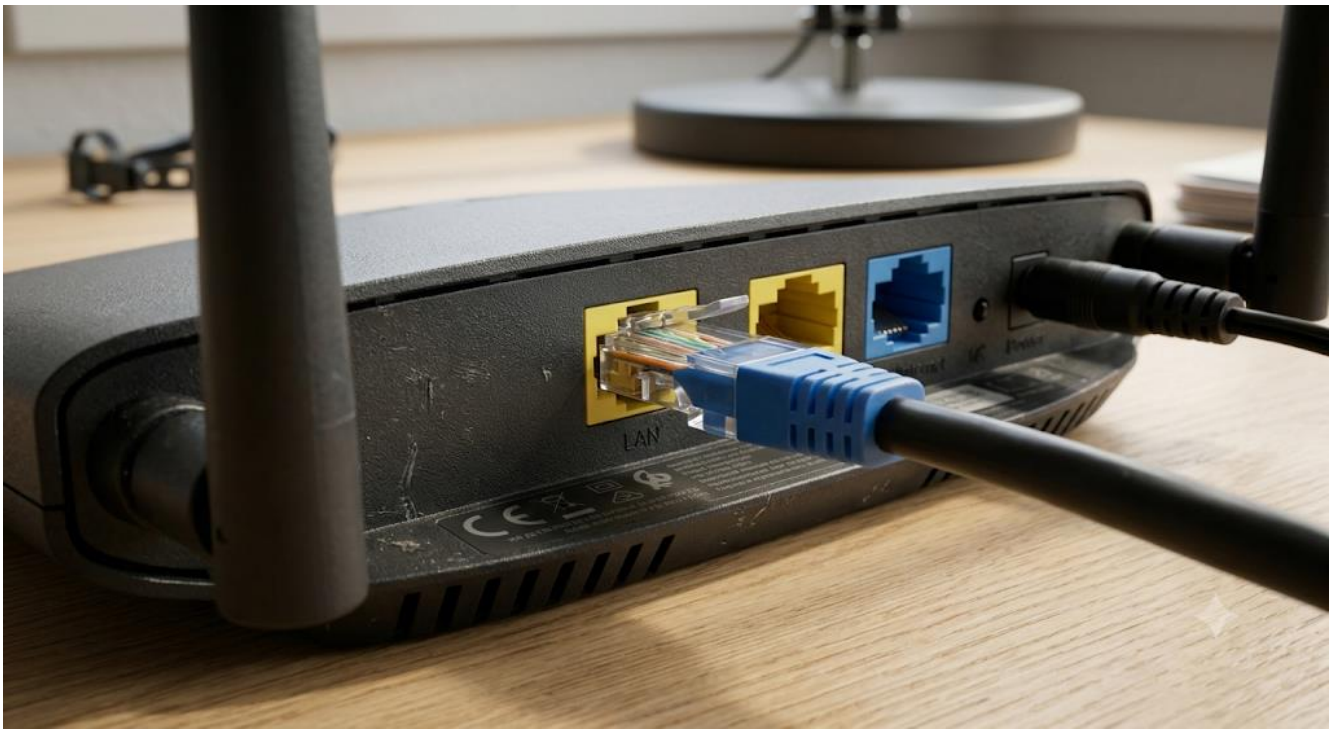


Figure 2: Gaming Zone Network Using Cat6 and RJ-45

Task 5: Network Cable / Connector Photo

For this task, add one real photo of a network cable or connector from home, college, a café, or another place. A photo showing an Ethernet cable connected to a router, switch or computer is suitable.



Caption:

Figure 3: Ethernet Cable with RJ-45 Connector Connected to Wi-Fi Router

Example description: This is an Ethernet cable with an RJ-45 connector. It is connected between the network device and a computer/router to provide a wired network connection.

Conclusion

In this assignment, I learned about Cat5e, Cat6 and fiber optic cables and the use of RJ-45 connectors. I also learned the T568B wire order and the basic steps used for crimping an

Ethernet cable. For a gaming zone, Cat6 with RJ-45 is a practical choice because it gives a stable and fast wired connection.

Quick Revision

- Cat5e = common Ethernet cable for basic networks.
- Cat6 = faster and reliable Ethernet connection.
- Fiber optic = very high speed and long-distance communication.
- RJ-45 = common connector used with Ethernet cables.
- T568B order = White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.